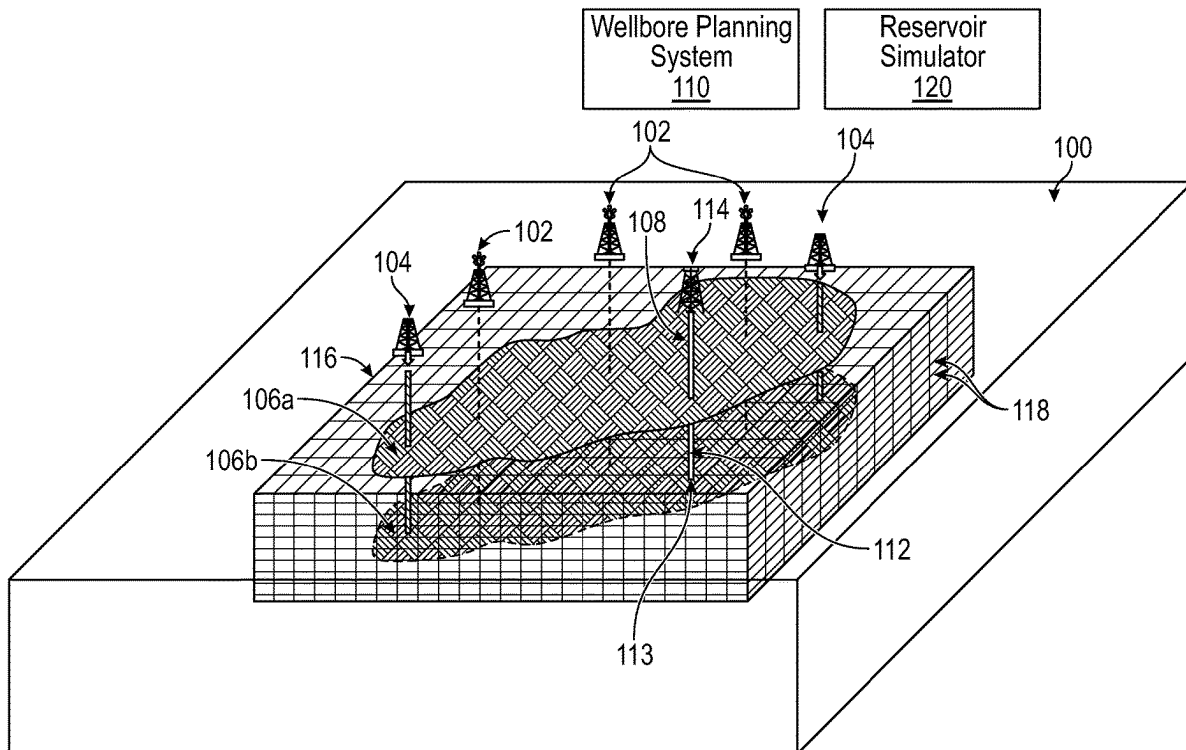




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Alramadhan(10) **Pub. No.: US 2025/0270923 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **METHODS AND SYSTEMS FOR
VALIDATION OF PERMEABILITY MODELS
BASED ON CUMULATIVE FLOW**(52) **U.S. Cl.**
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Dhahran (SA)(21) Appl. No.: **18/584,650**(22) Filed: **Feb. 22, 2024****Publication Classification**(51) **Int. Cl.**
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Examples of methods and systems are disclosed. The methods may include measuring, using flowmeter, a measured fluid flow rate at a plurality of depths in a well penetrating a hydrocarbon reservoir. The methods may also include, determining, using a reservoir simulator, a normalized cumulative fluid flow rate for the well based on the measured fluid flow rate at the plurality of depths, receiving a reservoir simulation model for the hydrocarbon reservoir, determining a normalized fluid flow capacity for the well based on the reservoir simulation model, and updating the reservoir simulation model based, at least in part, on the normalized cumulative fluid flow rate and the normalized fluid flow capacity.



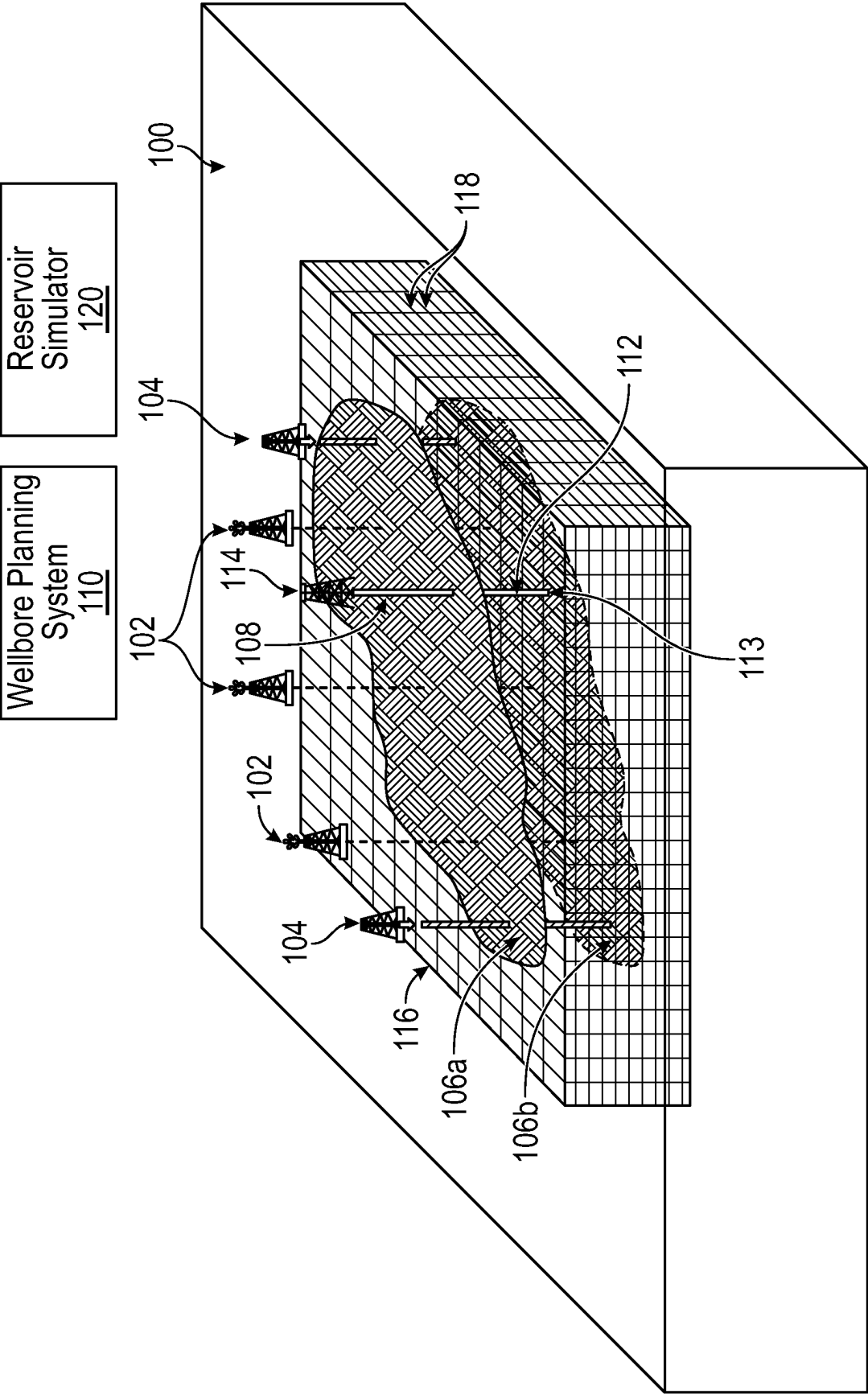


FIG. 1

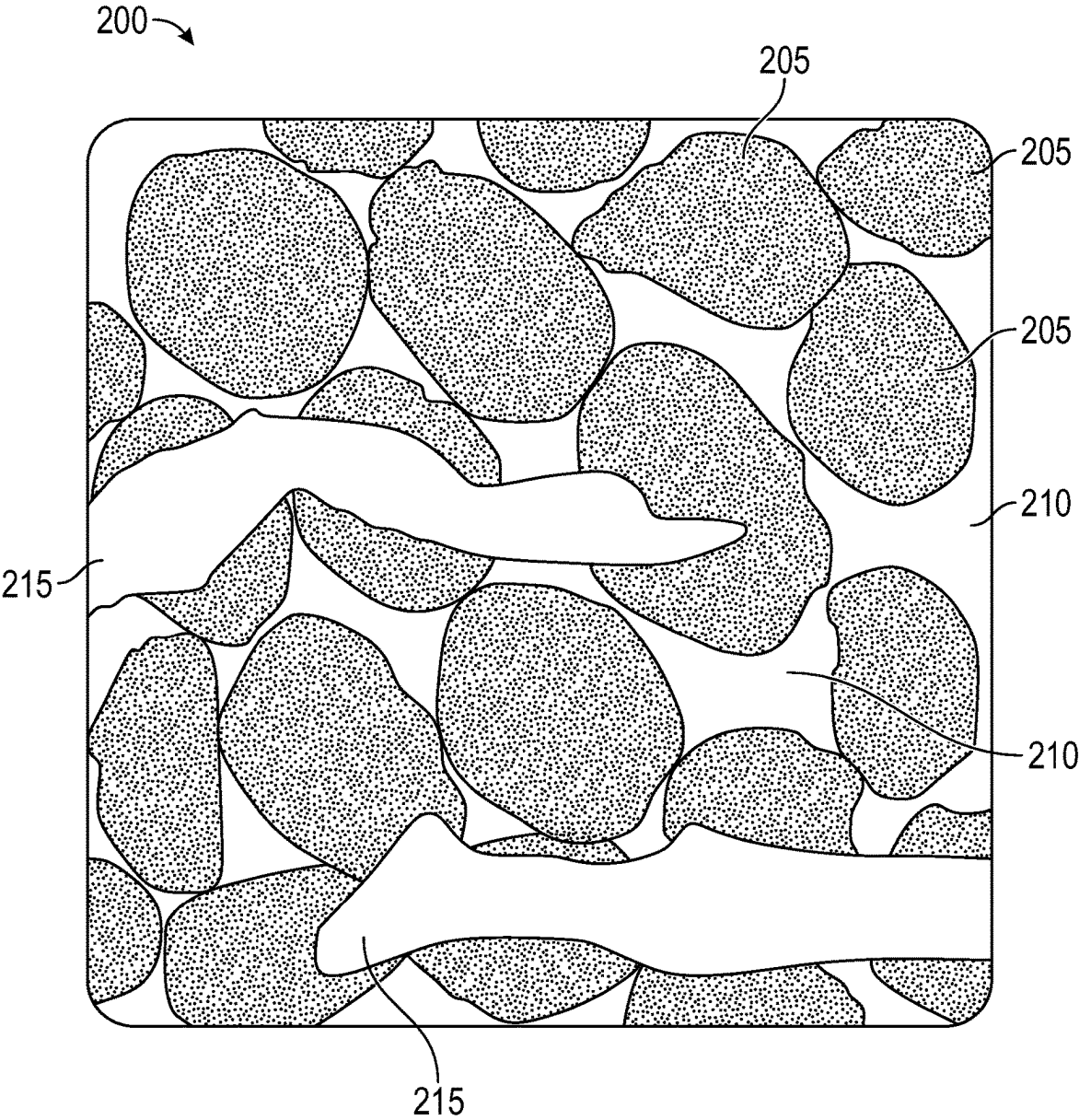


FIG. 2

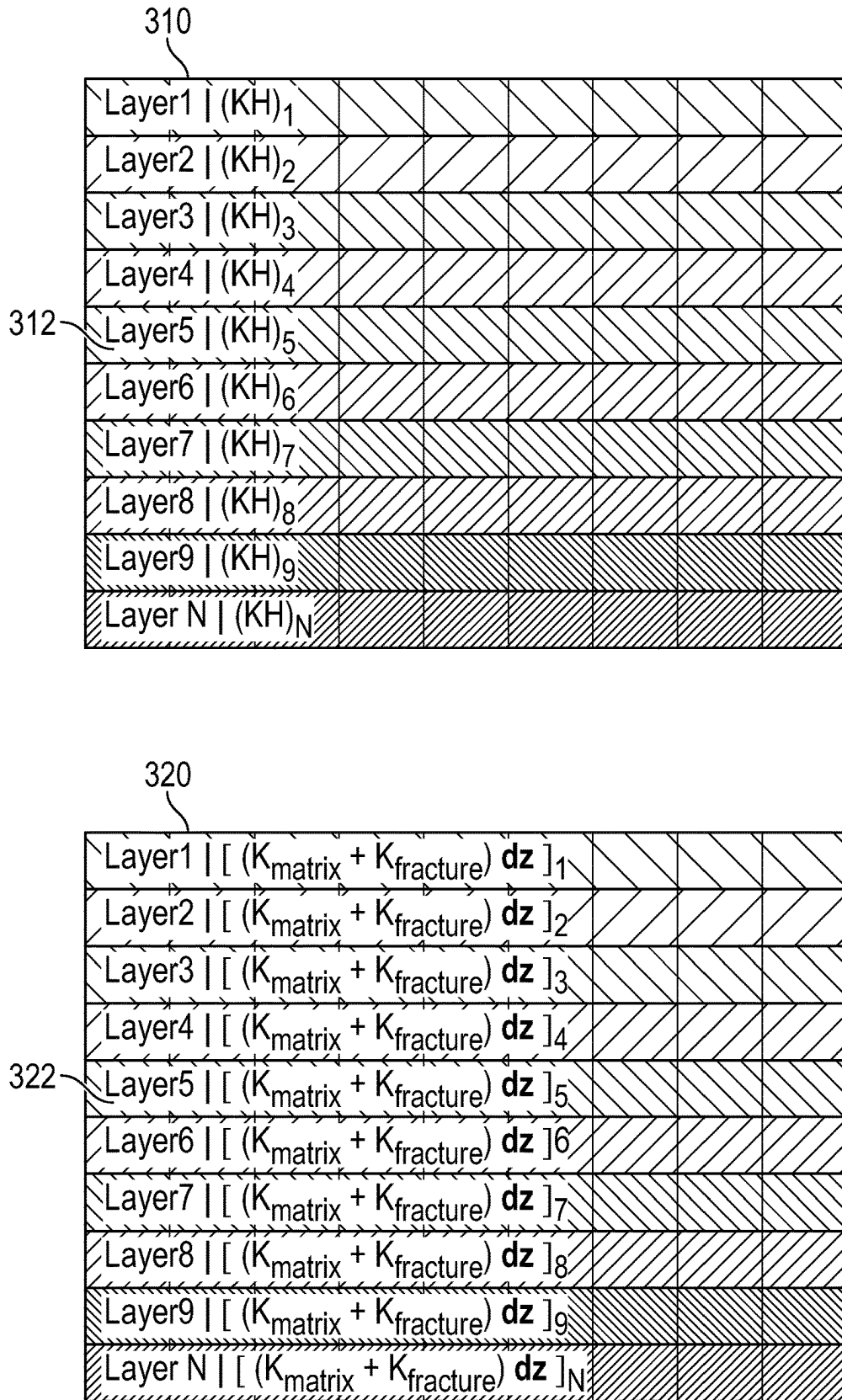


FIG. 3

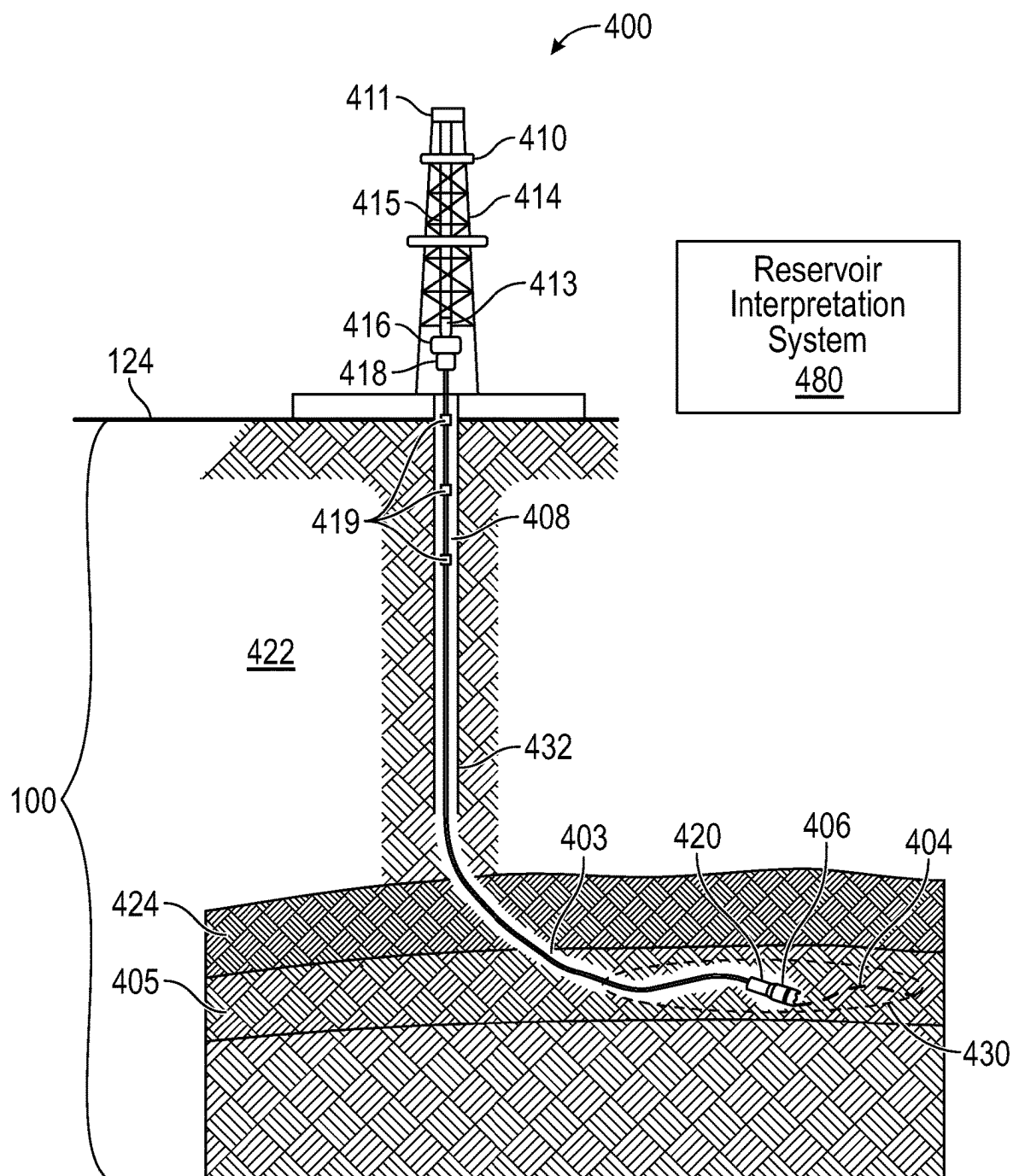


FIG. 4

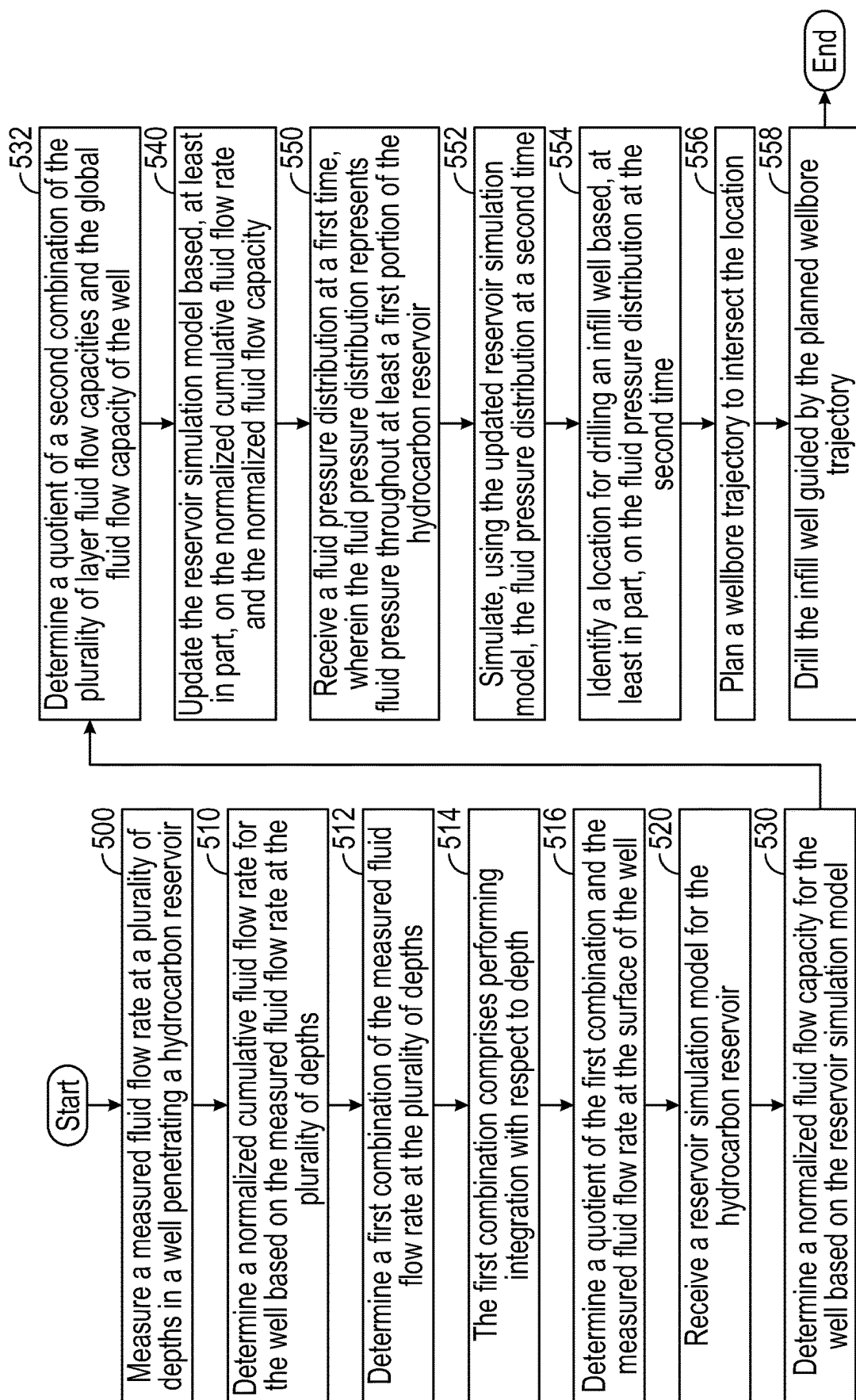


FIG. 5

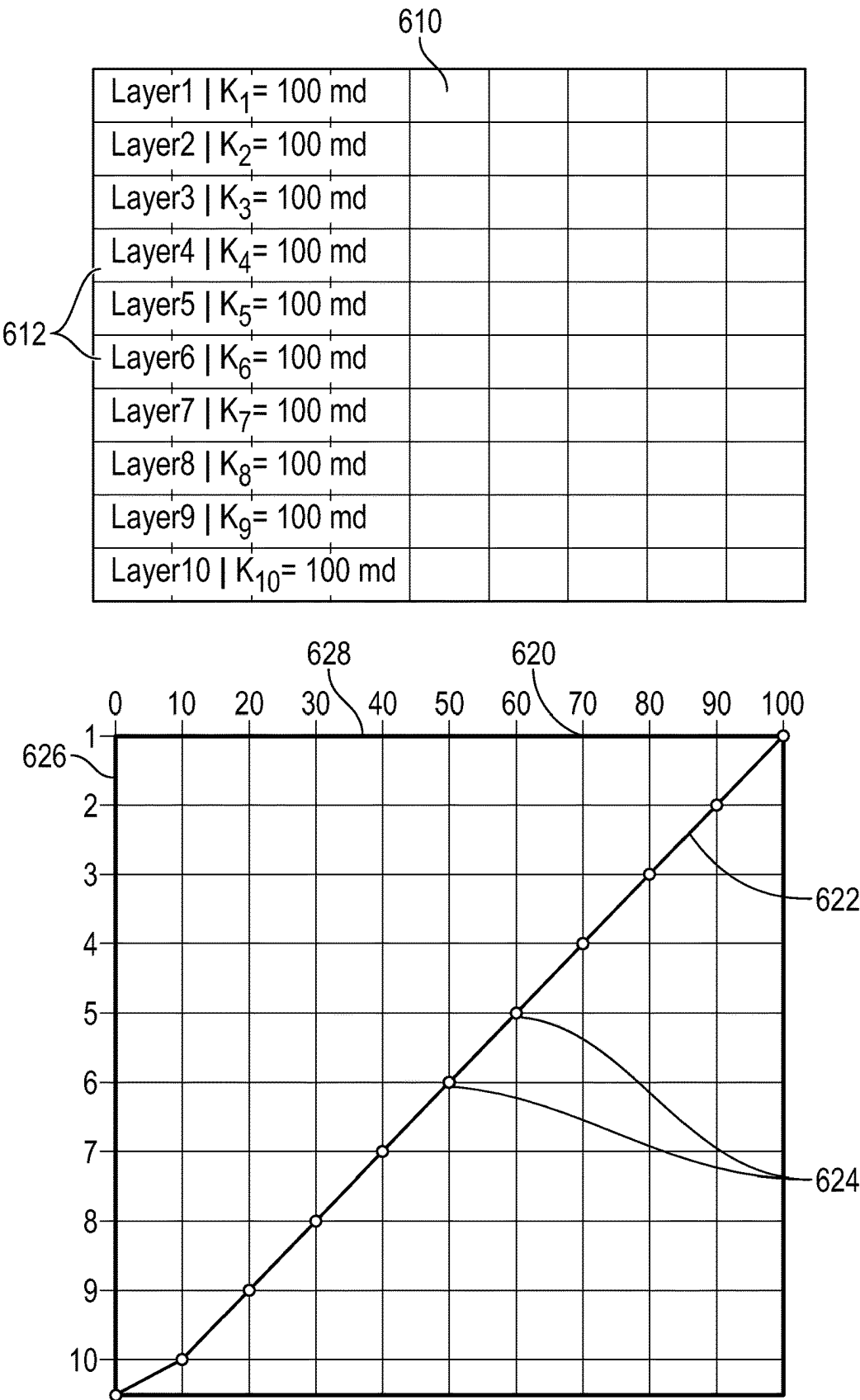


FIG. 6

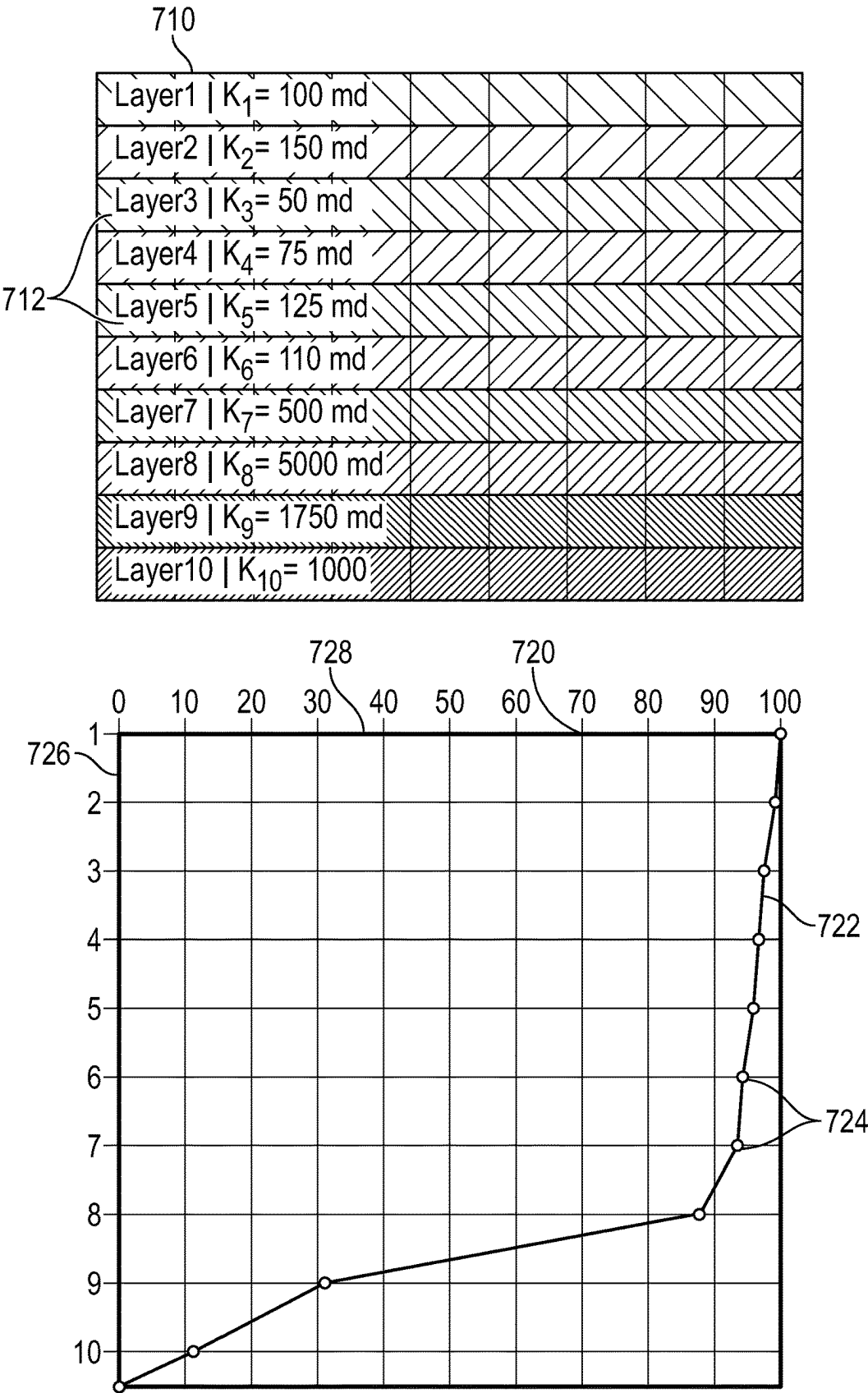


FIG. 7

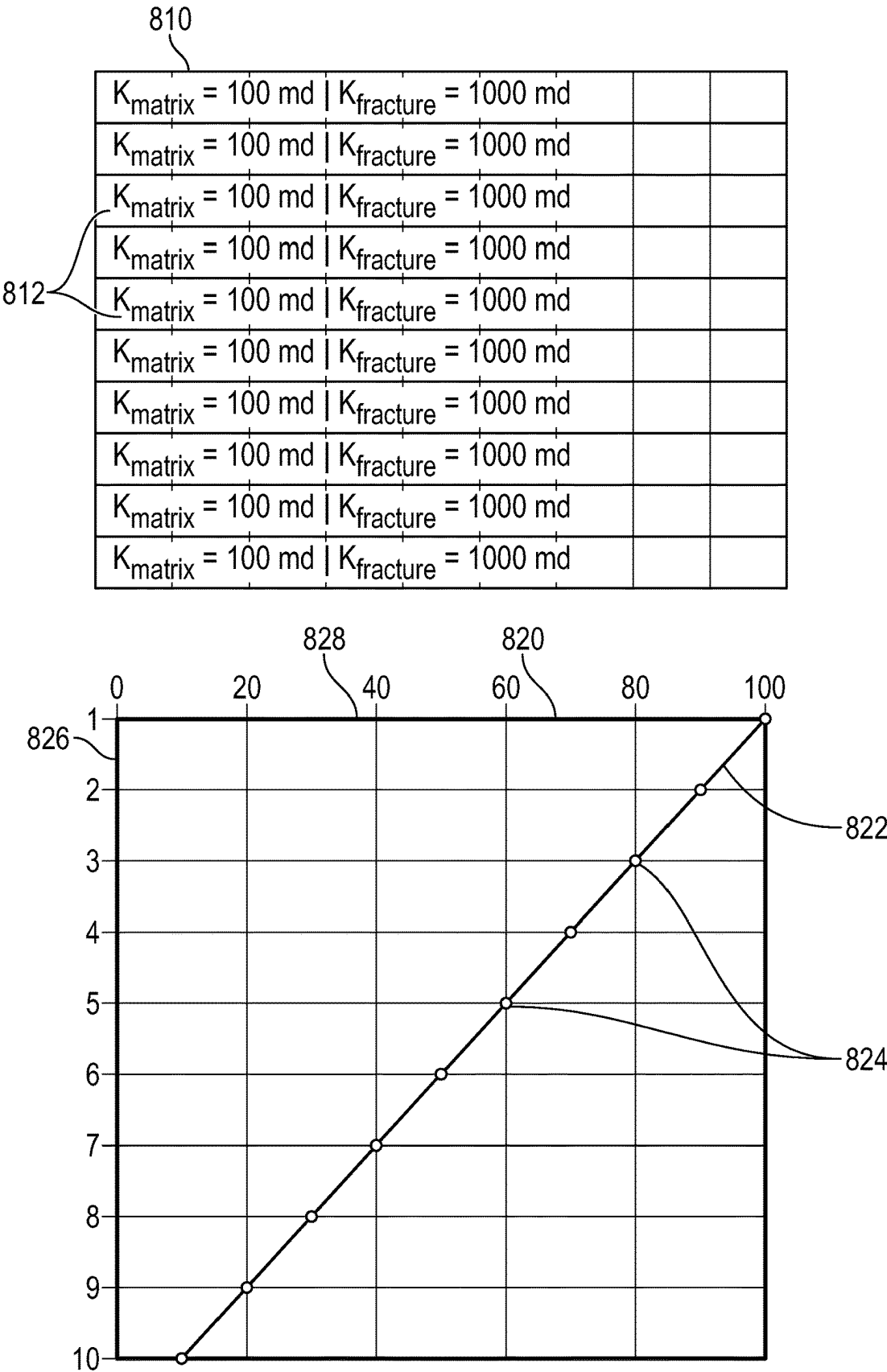


FIG. 8

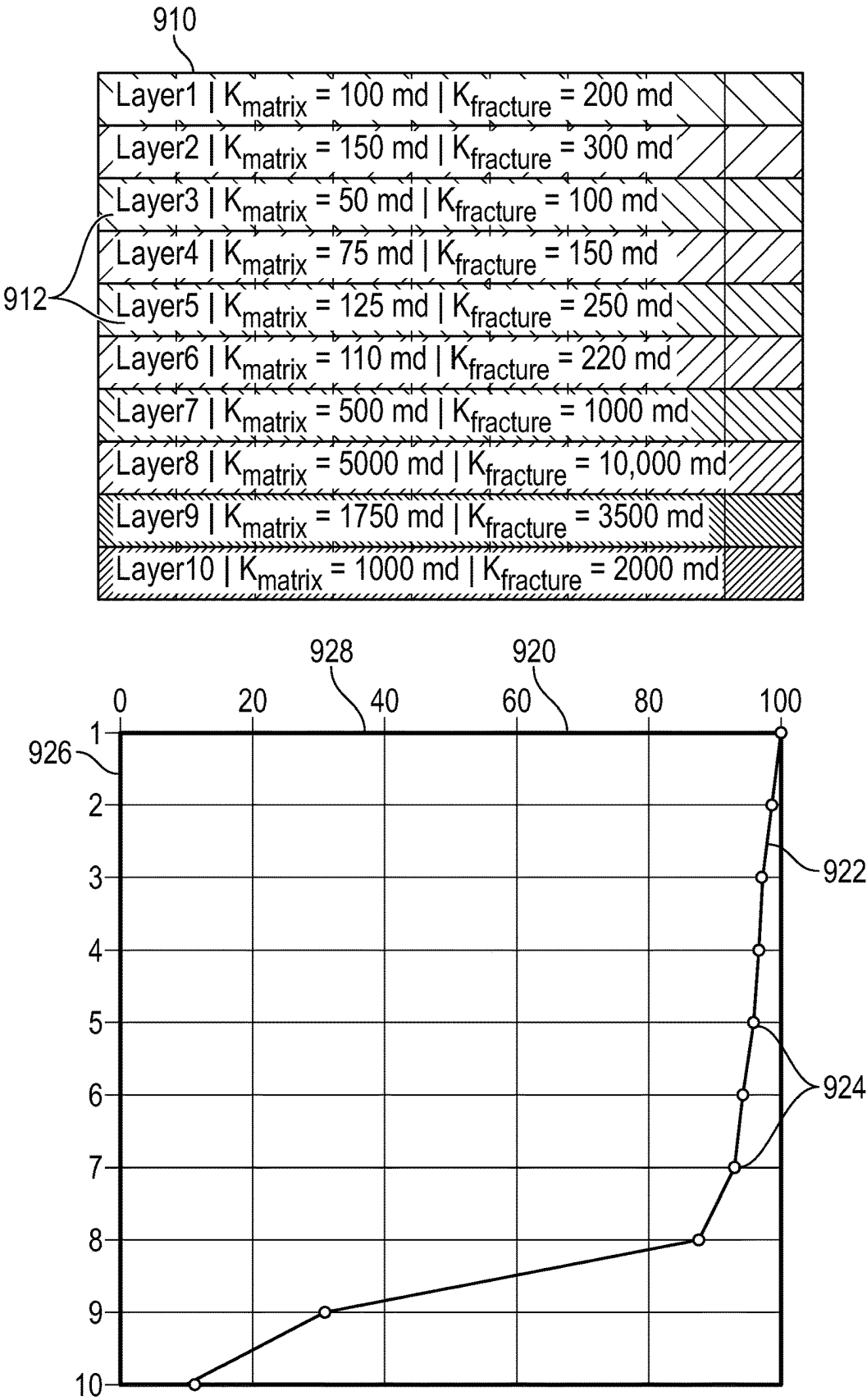


FIG. 9

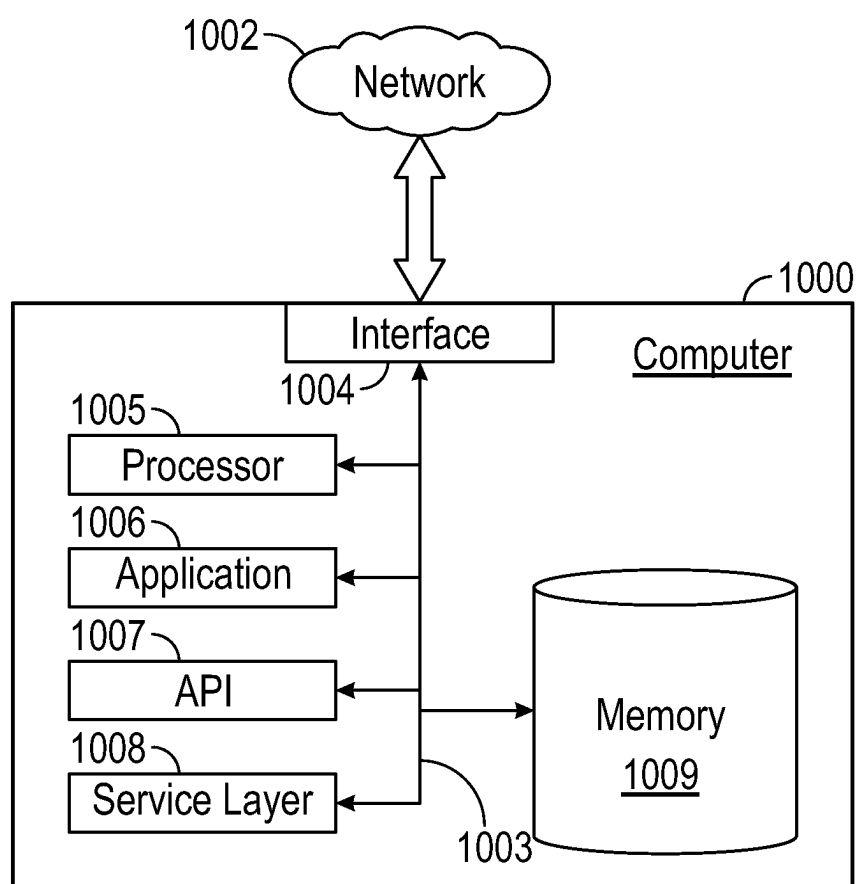


FIG. 10

METHODS AND SYSTEMS FOR VALIDATION OF PERMEABILITY MODELS BASED ON CUMULATIVE FLOW

BACKGROUND

[0001] Rocks may be composed of grains organized such that pores or space between the grains exist. The pores may be one mechanism that allows fluid, such as water and hydrocarbons, to flow through the rock. The ease with which fluid flows through the rock may be known as permeability. Another mechanism that allows fluid to flow through rock is fractures, such as natural fractures or man-made fractures. Quantifying rock permeability may be challenging, as well as expensive, due to the complex interplay between multi-modal pore systems of rocks and the heterogeneity and density of fractures that interrupt those pore systems.

[0002] However, it may be useful to collectively and individually quantify the matrix permeability and the fracture permeability of rocks. In turn, the matrix permeability and the fracture permeability may be used to predict dynamic fluid flow behavior in rock (i.e., a hydrocarbon production rate), inform reservoir simulation models, design completion strategies, evaluate the usefulness of recovery schemes, such as waterflooding and enhanced oil recovery, and determine future well placement.

SUMMARY

[0003] This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

[0004] In one aspect, embodiments disclosed herein relate to a method. The method includes measuring, using a flowmeter, a measured fluid flow rate at a plurality of depths in a well penetrating a hydrocarbon reservoir. The method also includes determining, using a reservoir simulator, a normalized cumulative fluid flow rate for the well based on the measured fluid flow rate at the plurality of depths, receiving a reservoir simulation model for the hydrocarbon reservoir, determining a normalized fluid flow capacity for the well based on the reservoir simulation model, and updating the reservoir simulation model based, at least in part, on the normalized cumulative fluid flow rate and the normalized fluid flow capacity.

[0005] In general, in one aspect, embodiments disclosed herein relate to a system. The system includes a flowmeter and a reservoir simulator. The flowmeter is configured to measure a measured fluid flow rate at a plurality of depths in a well penetrating a hydrocarbon reservoir. The reservoir simulator configured to receive the measured fluid flow rate at a plurality of depths, determine a normalized cumulative fluid flow rate for the well based on the measured fluid flow rate at the plurality of depths, receive a reservoir simulation model for the hydrocarbon reservoir, determine a normalized fluid flow capacity for the well based on the reservoir simulation model, and update the reservoir simulation model based, at least in part, on the normalized cumulative fluid flow rate and the normalized fluid flow capacity.

[0006] It is intended that the subject matter of any of the embodiments described herein may be combined with other embodiments described separately, except where otherwise contradictory.

[0007] Other aspects and advantages of the claimed subject matter will be apparent from the following description and the appended claims.

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 illustrates a subsurface region in accordance with one or more embodiments.

[0009] FIG. 2 shows examples of permeability in accordance with one or more embodiments.

[0010] FIG. 3 shows examples of reservoir simulation models in accordance with one or more embodiments.

[0011] FIG. 4 shows a drilling system in accordance with one or more embodiments.

[0012] FIG. 5 shows a flowchart in accordance with one or more embodiments.

[0013] FIG. 6 shows an example of a reservoir simulation model in accordance with one or more embodiments.

[0014] FIG. 7 shows an example of a reservoir simulation model in accordance with one or more embodiments.

[0015] FIG. 8 shows an example of a reservoir simulation model in accordance with one or more embodiments.

[0016] FIG. 9 shows an example of a reservoir simulation model in accordance with one or more embodiments.

[0017] FIG. 10 depicts a schematic diagram of a computer system in accordance with one or more embodiments.

DETAILED DESCRIPTION

[0018] In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.

[0019] Throughout the application, ordinal numbers (e.g., first, second, third, etc.) may be used as an adjective for an element (i.e., any noun in the application). The use of ordinal numbers is not to imply or create any particular ordering of the elements nor to limit any element to being only a single element unless expressly disclosed, such as using the terms “before”, “after”, “single”, and other such terminology. Rather, the use of ordinal numbers is to distinguish between the elements. By way of an example, a first element is distinct from a second element, and the first element may encompass more than one element and succeed (or precede) the second element in an ordering of elements.

[0020] In the following description of FIGS. 1-10, any component described regarding a figure, in various embodiments disclosed herein, may be equivalent to one or more like-named components described with regard to any other figure. For brevity, descriptions of these components will not be repeated regarding each figure. Thus, each and every embodiment of the components of each figure is incorporated by reference and assumed to be optionally present within every other figure having one or more like-named components. Additionally, in accordance with various embodiments disclosed herein, any description of the components of a figure is to be interpreted as an optional embodiment which may be implemented in addition to, in conjunction with, or in place of the embodiments described with regard to a corresponding like-named component in any other figure.

[0021] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a seismic signal” includes reference to one or more of such seismic signals.

[0022] Terms such as “approximately,” “substantially,” etc., mean that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

[0023] It is to be understood that one or more of the steps shown in the flowcharts may be omitted, repeated, and/or performed in a different order than the order shown. Accordingly, the scope disclosed herein should not be considered limited to the specific arrangement of steps shown in the flowcharts.

[0024] In general, disclosed embodiments include systems and methods to validate permeability models using measured flow rates at a well. In particular, some embodiments use an expected flowmeter response and a measured flowmeter response to validate the accuracy of permeability models. The measured flowmeter response may be converted to a dynamic parameter such as cumulative fluid flow, and the expected flowmeter response may be related to the permeability parameters of different layers traversed by the well. Combinations of the expected flowmeter response and a measured flowmeter response may be used to update or validate permeability models.

[0025] In the oil and gas industry, operations related to drilling and production in a wellbore involve collecting and processing a vast amount of information about the formation traversed by the wellbore. Information collected while drilling a wellbore may assist a driller in making decisions to optimize the drilling operation, for example, for maintaining or changing the direction of drill (geo-steering). Techniques to measure the conditions of the downhole drilling equipment, such as drill bit orientation, weight-on-bit, and torque are known as “measurement-while-drilling” (MWD) and techniques to measure the properties of the formation, such as resistivity, density, or sonic wave speed during drilling operations are known as “logging-while drilling” (LWD) techniques. The implementation of LWD techniques while a particular formation is being drilled provides information on physical properties of the formation and the changing conditions of the wellbore.

[0026] A formation must present certain physical characteristics to permit the storage and the flow of hydrocarbons. For example, a desired physical characteristic in a geomaterial of a formation is high porosity, that is, the presence of voids and spaces in the geomaterial. Similarly, high permeability, that is, the ease with a fluid can percolate through a geomaterial is desirable.

[0027] Reservoir simulation models play an important role in managing hydrocarbon resources in productive geological regions. In particular, a reservoir simulation model may be used to predict future oil, water, and gas production rates from one or more wells in an oil and/or gas reservoir. For example, the location of a new well site at the hydrocarbon reservoir and the design of production equipment for the well may be guided by reservoir simulations.

[0028] Reservoir simulation models are often modified to simulate production rates at wells with sufficient accuracy. In an initial stage, the reservoir simulation may be adjusted to match measured values of flow rates and/or downhole pressures. This stage is sometimes called “history-matching”. In a second predictive phase, the future behavior of the reservoir, including pore fluid pressure and phase changes of the fluid within the reservoir and production rates, may be simulated for various downhole pressure and production scenarios. Simulations may include cases where additional wellbore are drilled and/or fluid injection schedules are implemented. These predictions help inform the future operation of the reservoir.

[0029] Permeability is a parameter commonly used for reservoir simulation models and for predicting hydrocarbon production. Low matrix permeability and the presence of induced fractures often makes the accurate measurement of the matrix permeability a significant challenge for the oil and gas industry. The generation of matrix permeability or fracture permeability may rely on the judgement of the analyst, and may incur in errors, in particular in subsurface regions with complex geological structures. Methods and processing techniques to improve the validation of permeability in reservoir simulation models may assist in improving the accuracy and efficiency of reservoir simulations.

[0030] The resulting reservoir simulations may then be integrated into the established practical applications of searching for an extraction of hydrocarbon from subsurface hydrocarbon reservoirs. The disclosed methods represent an improvement over existing methods for at least the reasons of lower cost and increased efficacy.

[0031] FIG. 1 depicts a hydrocarbon reservoir in accordance with one or more embodiments. As illustrated in FIG. 1, a subsurface region (100) of interest may include one or more hydrocarbon reservoirs with various drilled wellbores such as, for example, production wells (102) or injection wells (104). As shown in FIG. 1, wells may extend from the surface of the Earth through various overburden layers to penetrate the hydrocarbon reservoir that may be bounded by an upper reservoir surface (106a) and a lower reservoir surface (106b). Furthermore, a wellbore planning system (110) may contemplate planned wells (108) (e.g., planned wellbore path (112)) at an identified hydrocarbon location (113) for the subsurface region (100). Wellbores may be drilled along a trajectory guided by the planned wellbore path (112) using a drilling system (114).

[0032] A reservoir simulation model (116) that corresponds to a portion of the subsurface region (100) may enclose at least a portion of the hydrocarbon reservoir. More specifically, the reservoir simulation model (116) may be represented by a plurality of grid blocks (118) that may be used to discretize reservoir properties, e.g., permeability, porosity or saturations. In some embodiments, the grid blocks may be cuboids, but in other embodiments the grid blocks may be irregularly shaped.

[0033] In some embodiments, a reservoir simulator (120) may generate one or more reservoir simulation models (116). For example, the reservoir simulator (120) may store well logs, fluid pressure distribution, and data regarding core samples for performing simulations. A reservoir simulator (120) may further analyze the well log data, the fluid pressure distribution, the core sample data, and other types of data to generate and/or update the one or more reservoir simulations models (116). In some embodiments, the reser-

voir simulator (120) may include a computer system that is similar to the computer system (1000) described below with regard to FIG. 10 and the accompanying description.

[0034] Turning to FIG. 2, FIG. 2 illustrates schematically a cross-section of a rock (200) in accordance with one or more embodiments. The rock (200) may be composed of grains (205). The grains (205) may be physically characterized by type, sphericity, roundness, roughness, connectivity, size, sorting of size, etc. Grain types include quartz, carbonate, or shale, for example. Such physical characteristics of the rock (200) may be described as the lithology of the rock (200).

[0035] The grains (205) may be organized such that pores (210) or space between the grains (205) exist. In turn, the lithology of the rock (200) may govern the types of pore systems that exist. Each rock (200) may contain complex, multi-model pore systems. Further, each pore system may overlap or invade another pore system. These pore systems may allow fluid, such as air, water, brine, hydrocarbons, or any combination thereof, to flow through the rock (200). The ease with which fluid flows through the rock (200) may be quantified as the permeability, a property of rock (200).

[0036] In some embodiments, rock permeability is modeled under the assumption of non-turbulent, Darcy-flow, and is considered to be scale dependent. As such, distinctions may be made for rock permeability related to fractures (215) (i.e., cracks or breaks) (macro-scale) and rock permeability related to pores (210) (micro-scale). Micro-scale permeability may be partially attributed to the random distribution of pores (210) throughout the rock. As such, these pore systems may govern the permeability of the rock (200). However, permeability may additionally be governed by the fractures (215) that interrupt the pore systems within the rock 200. Fractures (215) may be heterogeneous and densely pervade a rock (200). Heterogeneity may take the form of non-uniform apertures, roughness, and tortuosity. Further, fractures (215) may be natural or man-made. As such, quantifying permeability of rock (200) may be challenging due to the complex interplay between micro-scale permeability from pore systems and fracture permeability from heterogeneous fractures (215).

[0037] In some embodiments, micro-scale permeability is integrated in a reservoir simulation in the form of matrix permeability. In some embodiments, the matrix permeability may be generated from well data that may be collected using a well logging system. In other embodiments, the matrix permeability is determined from rock cores tested in a laboratory setting. Rock cores may be collected downhole using a rock coring system. Data obtained from rock core tests may be then corrected for depth and stress.

[0038] In some embodiments, measuring permeability in the laboratory includes encasing a core of known length and diameter in an air-tight sleeve. A fluid of known viscosity, such as, for example, nitrogen or brine, is injected into the core plug. The pressure drop across the sample and the flow rate are then measured. Permeability values are obtained by using Darcy's law. However, at low flow rates, a gas permeability is higher than the fluid permeability, because the gas does not adhere to the pore walls as liquid does. The slippage of gas along the pore walls gives rise to an apparent dependence of permeability on pressure, a phenomenon known as the Klinkenberg effect, and it is especially important in low-permeable rocks. Gas slippage may occur during measurements because nitrogen may be injected quickly,

and an equilibrium state may not be reached in a short time span. Therefore, permeability obtained from core measurements may need to be corrected due to the Klinkenberg effect.

[0039] In some embodiments matrix permeability may be obtained from un-cored wells, using data from petrophysical logs and building matrix permeability prediction models through correlations, artificial intelligence, machine learning techniques, and/or any other techniques for prediction modeling known in the art.

[0040] As discussed above, determining matrix permeability may involve complex measurement methodologies and simplifying modeling assumptions, thus the measurements of permeability values may be obtained with large uncertainties. Therefore, the quality of the matrix permeability measurements as well as permeability values obtained from petrophysical logs is commonly assessed in petrophysical applications. For example, the quality of a matrix permeability model may be evaluated using a histogram of error, defined by the expression:

$$K_{model} - K_{core} \quad \text{Equation (1)}$$

where K_{model} represents predicted permeability values (for example, from petrophysical logs) and K_{core} represents directly measured permeability values (for example, from core samples). A matrix permeability model with a histogram of error centered around zero is a favorable model because it indicates that the matrix permeability model neither systematically overpredicts nor systematically underpredicts measured values. Another example of an indicator used to evaluate the quality of matrix permeability model is the root-mean-square (rms) value of the cross plot K_{model} vs. K_{core} . A high rms value results from a strong correlation between K_{model} and K_{core} , and thus it indicates a high quality of the matrix permeability model. However, these examples of indicators used to validate the matrix permeability model are based on data obtained from laboratory measurements, which may be limited and/or difficult to obtain. Methods and systems to validate permeability models using data measured independently from the data used to generate the models may assist in the construction of more reliable permeability models.

[0041] In some embodiments, fracture permeability is based on in situ measurements with injection flow rate tests used to identify areas of high permeability associated with fractured zones. In other embodiments, fracture permeability may be obtained using a geometric measuring tool to generate an image of the fractured rock. The output image may be processed using a computer program to distinguish the rock from the voids in order to calculate the aperture of the fracture. Fracture permeability may then be directly estimated from the aperture of the fracture. For example, fracture aperture values may be estimated from borehole image logs using the following equation:

$$w = cAR_m^b R_{xo}^{1-b} \quad \text{Equation (2)}$$

where w is the aperture of the fracture, c and b are tool-dependent constants, A is the excess current generated when

an array of buttons are close to fractures, R_m^b is mud resistivity, and R_{xo}^{1-b} is the flushed zone resistivity taken from calibrated images. In some embodiments, fracture permeability provides a second component of rock permeability, which, when combined with the matrix permeability, may form the basis for permeability inputs for a reservoir simulation model (116).

[0042] In some embodiments, a reservoir simulator (120) may make use of the matrix permeability and/or the fracture permeability to simulate the fluid flow rate of a production well (102) using a reservoir simulation model (116). The matrix permeability in single-porosity rocks, and fracture permeability in the case of fractured-reservoir rocks, may be used as inputs into single-porosity and dual-porosity dual-permeability reservoir simulation models (116), respectively. For single-porosity reservoir simulation models, the matrix permeability may be the controlling parameter of the simulated productivity index of wells. For dual-porosity dual-permeability reservoir simulation models, the combined matrix permeability and fracture permeability may be the controlling parameters of the simulated productivity index of wells. In some embodiments, the matrix permeability and/or the fracture permeability may be used in multi-phase flow simulations to history match historical multi-phase flow production or to predict future multi-phase flow production. In view of the influence that the matrix permeability and/or fracture permeability may have on predicted well production, their verification with actual measured data not used in the construction of the permeability models, may lead to more accurate reservoir simulations used to forecast well production.

[0043] Returning to FIG. 1, in reservoir simulations, the plurality of grid blocks (118) may refer to an original cell of a reservoir simulation model (116) as well as coarse grid blocks that may refer to an amalgamation of original cells of the reservoir simulation model (116). For example, a grid cell may be the case of a 1x1 grid block, where coarse grid blocks may be of sizes 2x2, 4x4, 8x8, etc. Both the grid cells and the coarse grid blocks may correspond to columns for multiple model layers within the reservoir simulation model (116).

[0044] Flow properties, such as flux, may be defined as a reservoir fluid (e.g., oil or natural-gas) flowing between any two grid blocks (118). Likewise, grid cells or blocks may be upscaled in a method that reduces the computational demand on running simulations using fewer grid cells. However, a grid model may lose accuracy in a reservoir simulation if the underlying properties differ too much from the original reservoir simulation model (116).

[0045] In some embodiments, a fluid production rate may be determined for one or more production wells (102) in the subsurface region (100) using the reservoir simulation model (116). A bottom-hole pressure value may be determined using the production rate of each production well (102). For example, a measured flowing bottom-hole pressure value, P_{wf} , may be related to simulated grid pressures of fluid flow using the following equation, also known as Peaceman well equation:

$$P_{wf} = P_o + \frac{q\mu}{2\pi k\Delta z} \ln\left(\frac{r_w}{r_o}\right) \quad \text{Equation (3)}$$

where P_o is simulated well-block pressure, μ is viscosity, k is permeability, Δz is the thickness of the grid block (118), q is the flow rate per thickness Δz , r_w is wellbore radius, and r_o is the equivalent radius of the well block. For a medium that is isotropic and discretized using cubic grid blocks, the following values may be adopted for r_o :

$$r_o = 0.2(\Delta x) \quad \text{Equation (4)}$$

where Δx is grid spacing in x-direction. In some embodiments the permeability k corresponds to the matrix permeability characteristic of a single-flow model. In other embodiments, the permeability k corresponds to a combination of the matrix permeability and the fracture permeability characteristic of a dual-porosity, dual-permeability model.

[0046] Peaceman's well equation, as given by Eq. (3), may be rearranged to solve for the flow rate q per thickness Δz :

$$q = \frac{2\pi k(P_{wf} - P_o)\Delta z}{\mu \ln\left(\frac{r_o}{r_w}\right)} \quad \text{Equation (5)}$$

Integrating both sides of Eq. (5) with respect to depth over the thickness of the of the grid block (118) gives the expression:

$$\int_{Z_o}^Z q dz = \int_{Z_o}^Z \frac{2\pi k(P_{wf} - P_o)}{\mu \ln\left(\frac{r_o}{r_w}\right)} dz \quad \text{Equation (6)}$$

where the integral limits Z_o and Z are respectively the deepest and shallowest depth of wellbore area open to flow within the grid block (118). Under steady-state flow, the pressures P_{wf} and P_o , as well as the viscosity μ are constant, and therefore P_{wf} , P_o , μ can be taken out of the integrand along with the radii r_o and r_w , as follows:

$$\int_{Z_o}^Z q dz = \frac{2\pi(P_{wf} - P_o)}{\mu \ln\left(\frac{r_o}{r_w}\right)} \int_{Z_o}^Z k dz \quad \text{Equation (7)}$$

The integral on the right hand side in Equation (7) may be interpreted as the cumulative flow along the depth portion, or total thickness, $H_w = (Z - Z_o)$ of the wellbore. The cumulative flow provides the total amount of fluid flowing through the wellbore in a period of time and may be used to monitor well production.

[0047] In some embodiments, the total thickness $H_w = (Z - Z_o)$ maybe be discretized into a number N of finite layers. Each layer may be characterized by a flow rate q_i and a flow-capacity Q_i . FIG. 3 show examples of finite layers used to discretize $(Z - Z_o)$. Panel (310) illustrates an example of 10 layers (312) used to discretize $(Z - Z_o)$ for a single-flow model. In the example of a single-flow model, the flow capacity of the i -th layer (312) may be given by the product of the layer permeability k_i and layer thickness H_i , namely, $Q_i = k_i H_i$. On the other hand, panel (320) shows an example

of 10 layers (322) used to discretize $(Z-Z_o)$ for a dual-porosity, dual-permeability model. The flow capacity of the i -th layer (322) in Panel (320) may then be given by the product:

$$Q_i = (k_{matrix} + k_{fracture})_i H_i \quad \text{Equation (8)}$$

[0048] Furthermore, the flow rate q_i and flow capacity of the i -th layer (312, 322) may be related to the flow rate of the well q_w and the flow capacity of the well Q_w by Darcy law:

$$\frac{q_i}{Q_i} = \frac{q_w}{Q_w} \quad \text{Equation (9)}$$

where q_w may be a flow rate of the well measured at the surface. As a non-limiting example, q_w may be measured at the surface during acquisition of a flowmeter response. In addition, the flow capacity of the well Q_w may be obtained with the product of the total thickness $H_w=(Z-Z_o)$ and the permeability corresponding to the total thickness K_w , $Q_w=K_w H_w$.

[0049] In some embodiments, the flowmeter response is represented as a normalized cumulative flow as a function of wellbore depth. If the flow rate of the well q_w is used for normalization, the normalized cumulative flow may be expressed as follows:

$$\frac{\int_{Z_o}^Z q dz}{q_w} \quad \text{Equation (10)}$$

Discretizing with a number N of finite layers with flow rates q_i , the integral in Equation (10) may be simplified to:

$$\int_{Z_o}^Z q dz = \sum_{i=1}^N q_i \quad \text{Equation (11)}$$

Therefore, combining Equations (9) and (11) the normalized cumulative flow provided by the flowmeter may be related to the flow capacities of the layers traversed by the wellbore:

$$\frac{\int_{Z_o}^Z q dz}{q_w} = \frac{\sum_{i=1}^N Q_i}{Q_w} \quad \text{Equation (12)}$$

The right-hand side of Equation (12) may then be interpreted as the normalized fluid flow capacity of the well. The right-hand side of Equation (12) may be also interpreted as the expected flowmeter response of a vertical well for a given permeability profile along the wellbore, while the left-hand side may be interpreted as the normalized cumulative flow measured at the well surface. Furthermore, since the flow capacities of the layers Q_i and the total flow capacity Q_w may be directly obtained from the permeability model as discussed above, Equation (12) may allow the

validation of a permeability model with actual measurements of the normalized cumulative flow.

[0050] In particular, when a production well (102) is active, the production well (102) is acquiring hydrocarbon production from a hydrocarbon reservoir, and the fluid flow rate may be measured. Furthermore, where production data may exist for an active production well, such production data may be used to validate parameters of a reservoir simulation model (116), such as, the permeability matrix and/or the fracture permeability. The reservoir simulation model (116) may then be updated for achieving an improved match between simulated production and measured production. The reservoir simulation model (116) updated in this manner may be considered to be an accurate representation of the actual hydrocarbon reservoir. Using the updated reservoir simulation model (116) the reservoir simulator (120) may simulate future production for various scenarios, for example, injection schedules, future proposed wells, and artificial lift programs, among others.

[0051] FIG. 4 shows a drilling system (400) in accordance with one or more embodiments. As shown in FIG. 4, a wellbore (403) following a wellbore trajectory (404) may be drilled by a drill bit (406) attached by a drillstring (408) to a drilling rig (410) located on the surface (124) of the earth. The drilling rig (410) may include framework, such as a derrick (414) to hold drilling machinery. A crown block (411) may be mounted at the top of the derrick (414), and a traveling block (413) may hang down from the crown block (411) by means of a cable (415) or drilling line. One end of the cable (415) may be connected to a drawworks (not shown), which is a reeling device that may be used to adjust the length of the cable (415) so that the traveling block (413) may move up or down the derrick (414).

[0052] A top drive (416) provides clockwise torque via the drive shaft (418) to the drillstring (408) in order to drill the wellbore (403). The drillstring (408) may comprise a plurality of sections of drill pipe attached at the uphole end to the drive shaft (418) and downhole to a bottomhole assembly ("BHA") (420). The BHA (420) may be composed of a plurality of sections of heavier drill pipe and one or more measurement-while-drilling ("MWD") tools configured to measure drilling parameters, and one or more logging tools. The drilling parameters may include, without limitation, torque, weight-on-bit, drilling direction, temperature, etc. The logging tools may be configured to measure parameters of the rock surrounding the wellbore (403), such as electrical resistivity, density, sonic propagation velocities, gamma-ray emission, etc. MWD and logging tools may include sensors and hardware to perform measurements, and these measurements may be transmitted to the surface (124) using any suitable telemetry system known in the art. The BHA (420) and the drillstring (408) may include other drilling tools known in the art but not specifically shown.

[0053] The wellbore (403) may traverse a plurality of overburden (422) layers and one or more formations (424) to a hydrocarbon reservoir (405) within the subterranean region of interest (100), and specifically to a drilling target (430) within the hydrocarbon reservoir (405). The wellbore trajectory (404) may be a curved or a straight trajectory. All or part of the wellbore trajectory (404) may be vertical, and some parts of the wellbore trajectory (404) may be deviated or have horizontal sections. One or more portions of the wellbore (403) may be cased with casing (432) in accordance with a wellbore plan.

[0054] To start drilling, or “spudding in” the well, the hoisting system lowers the drillstring (408) suspended from the derrick (414) towards the planned surface location of the wellbore (403). An engine, such as an electric motor, may be used to supply power to the top drive (416) to rotate the drillstring (408) through the drive shaft (418). The weight of the drillstring (408) combined with the rotational motion enables the drill bit (406) to bore the wellbore (403).

[0055] In one or more embodiments, the drilling system (400) may receive well-measured data from one or more sensors and/or logging tools arranged to measure controllable parameters of the drilling operation. During operation of the drilling system (400), the well-measured data may include mud properties, flow rates, drill volume and penetration rates, rock physical properties, etc.

[0056] In particular, flow rates may be measured with one or more flowmeters (419), or any other device installed at the wellbore (403) to measure fluid flow rate. Flowmeters (419) can be used to measure flow rates of liquids or gases in-situ. Non-limiting examples of flowmeters include the spinner flowmeter, the torque flowmeter, the cross correlation flowmeter, the differential-pressure meter, the orifice meter, the positive-displacement meter, the vortex meter and the multiphase meter. The spinner and torque flowmeters measure the average velocity of the fluids crossing the flowmeter, while the cross-correlation flowmeter measures the velocity of a particular phase. Other possible techniques to measure flow rates known to those skilled in the art may be implemented without departing from the scope of the present disclosure.

[0057] The drilling system (400) may be disposed at and communicate with other systems in the well environment, such as a reservoir interpretation system (480) and a wellbore planning system (110). The drilling system (400) may control at least a portion of a drilling operation by providing controls to various components of the drilling operation.

[0058] A reservoir interpretation system (480) is primarily used by geoscientists, seismic interpreters, and exploration teams in the oil and gas industry for analyzing data to understand subsurface geological structures. Reservoir modelers use the reservoir interpretation system (480) to visualize data to provide insights into subsurface structures, faults, and potential hydrocarbon reservoirs.

[0059] A reservoir interpretation system (480) may enable interpreters to identify and interpret structures that may impact hydrocarbon reservoirs. Data analysis tools in the reservoir interpretation system (480) aid in defining reservoir characteristics, identifying anomalies, and highlighting potential hydrocarbon traps. Interpreters may use the reservoir interpretation system (480) to analyze and characterize hydrocarbon reservoirs by integrating different data sources, including seismic data, well logs, production data, and seismic inversion results. Workstations provide tools for reservoir property estimation, quantitative analysis, and reservoir performance evaluation. The reservoir characteristics may help in optimizing well locations and predicting hydrocarbon distribution.

[0060] The reservoir interpretation system (480) may facilitate prospect generation and evaluation, where interpreters identify and assess areas with high hydrocarbon exploration potential. They can perform detailed geological and geophysical analysis, identify drilling targets, and quantify the risk and uncertainty associated with potential prospects. Finally, workstations enable interpreters to collabo-

rate with team members, share interpretation results, and communicate findings effectively. Interpretation software allows for the creation of reports, annotated images, and presentations to communicate geological interpretations to stakeholders.

[0061] The reservoir interpretation system (480) may assist geoscientists involved in exploration and production activities, helping them make informed decisions about drilling locations, optimize production strategies, and understand complex subsurface geological structures. The reservoir interpretation system (480) may be a specialized computer system used by geoscientists and interpreters for analyzing and interpreting seismic data. The reservoir interpretation system (480) may be implemented on a computing device such as that shown in FIG. 10.

[0062] A high-performance reservoir interpretation system (480) with a powerful processor, ample memory, and a high-resolution display is essential to handle computationally demanding tasks efficiently. Dedicated GPUs may be crucial for real-time rendering of high-volumes of data, enabling smooth and interactive visualization. GPUs with high memory and parallel processing capabilities accelerate tasks like volume rendering and horizon visualization.

[0063] Reservoir interpretation often involves working with large and complex datasets. Multiple high-resolution monitors allow interpreters to view seismic data, cross-sections, time slices, attribute maps, and other visualizations simultaneously, enhancing productivity and analysis accuracy. The reservoir interpretation system (480) may be equipped with industry-standard software applications tailored for data interpretation, data processing and visualization tools, geological structure interpretation systems, and 3D modeling software. A high-capacity and fast storage system, such as solid-state drives (SSDs) or RAID arrays, is necessary to store and access complex datasets efficiently. The reservoir interpretation system (480) often requires network connectivity to access centralized data repositories, collaborate with colleagues, and share interpretation results. A robust network infrastructure with fast Ethernet or fiber connections ensures smooth data transfer and collaboration capabilities.

[0064] Essential peripherals like keyboards, mice, and graphics tablets enable efficient interaction with data and software interfaces. Additionally, color-calibrated and high-accuracy input devices enhance the precision of interpretation tasks like picking horizons or drawing geological features. The reservoir interpretation system (480) should have backup solutions in place to protect valuable data from loss or damage. Automated backup systems, external storage devices, or network-attached storage (NAS) can be utilized to ensure data safety. In some cases, interpreters may need remote access to the reservoir interpretation system (480) or collaborate with colleagues remotely. Setting up remote access capabilities, such as Virtual Private Networks (VPNs) or remote desktop solutions, allows interpreters to work from different locations and share their work effectively. The reservoir interpretation system (480) may be customized to meet the needs of interpreters and the specific requirements of projects. The hardware specifications may vary based on factors like the complexity of interpretations, the size of datasets, and the software tools utilized.

[0065] In some embodiments, reservoir simulations may be used by the reservoir interpretation system (480) to help determine a location of a hydrocarbon reservoir (405) (or

other subterranean features). Knowledge of the existence and location of the hydrocarbon reservoir (405) and other subterranean features may be transferred from the reservoir interpretation system (480) to a wellbore planning system (110). The wellbore planning system (110) may use information regarding the hydrocarbon reservoir (405) location to plan a well, including a wellbore trajectory (404) from the surface (124) of the earth to penetrate the hydrocarbon reservoir (405). In addition, to the depth and geographic location of the hydrocarbon reservoir (405), the planned wellbore trajectory (404) may be constrained by surface limitations, such as suitable locations for the surface position of the wellhead, i.e., the location of potential or preexisting drilling rigs, drilling ships or from a natural or man-made island.

[0066] Typically, the wellbore plan is generated based on best available information at the time of planning from a geophysical model, geomechanical models encapsulating subterranean stress conditions, the trajectory of any existing wellbores (which it may be desirable to avoid), and the existence of other drilling hazards, such as shallow gas pockets, over-pressure zones, and active fault planes. Information regarding the planned wellbore trajectory (404) may be transferred to the drilling system (400) described in FIG. 4. The drilling system (400) may drill the wellbore (403) along the planned wellbore trajectory (404) to access the drilling target (430) in the hydrocarbon reservoir (405).

[0067] The wellbore planning system (110) is used in the oil and gas industry for designing and planning drilling operations. It assists drilling engineers and teams in making strategic decisions related to wellbore placement, casing design, trajectory planning, and well path optimization. The wellbore planning system (110) allows drilling engineers to visualize and interact with wellbore data in a 3D environment. It provides a graphical representation of the planned well trajectory, existing well paths, geological formations, and potential hazards.

[0068] The wellbore planning system (110) integrates geological models, well logs, seismic data, and other subsurface information to facilitate the creation of accurate and realistic wellbore plans. By incorporating geological models, drilling engineers can optimize well placement in reservoir targets and avoid geohazards. Furthermore, the wellbore planning system (110) may assist in designing optimal well trajectories based on reservoir targets, geologic constraints, and drilling objectives. Engineers can define well paths that maximize drilling efficiency, reach specific targets (horizontal or vertical), and account for geological formations and structural complexities.

[0069] The wellbore planning system (110) incorporates collision-avoidance algorithms to assess potential collision risks between nearby wells, salt bodies, or other subsurface infrastructure. By considering uncertainties in subsurface data and drilling conditions, the wellbore planning system (110) may assess collision probabilities for planned well paths. This analysis helps in quantifying risks associated with collision potential and improving well placement decisions. The wellbore planning system (110) provides real-time alerts to prevent wellbore collisions and maintain drilling safety.

[0070] The wellbore planning system (110) helps drilling engineers in designing casing strings and selecting appropriate tubulars based on the wellbore conditions, planned drilling operations, and regulatory requirements. It considers

factors such as pressure, temperature, well depth, formation properties, and casing load capacity. Furthermore, the wellbore planning system (110) performs torque and drag analysis to evaluate the forces and stresses acting on the drillstring during drilling operations. This analysis helps in identifying potential issues such as differential sticking, buckling, or limitations in the drilling equipment. The wellbore planning system (110) may have the capability to integrate real-time drilling data, such as downhole measurements, drilling parameters, and formation evaluation results. This integration allows engineers to monitor the drilling progress, make on-the-fly adjustments to the well plan, and optimize drilling efficiency. Furthermore, the wellbore planning system (110) provides tools for generating reports, exporting data, and documenting drilling plans and decisions. These reports can be shared with regulatory agencies, drilling contractors, and other stakeholders to ensure alignment and compliance throughout the drilling lifecycle.

[0071] The wellbore planning system (110) assists drilling engineers in designing optimal well trajectories, minimizing risks, and maximizing drilling efficiency. They integrate various subsurface data sources, perform complex analyses, and provide visualization tools to support informed decision-making in well planning and drilling operations.

[0072] Turning to FIG. 5, FIG. 5 shows a flowchart in accordance with one or more embodiments. Specifically, FIG. 5 describes an embodiment of the inventive method to validate permeability models from measured cumulative flow rates. While the various blocks in FIG. 5 are presented and described sequentially, one of ordinary skill in the art will appreciate that some or all of the blocks may be executed in different orders, may be combined or omitted, and some or all of the blocks may be executed in parallel. Furthermore, the blocks may be performed actively or passively.

[0073] The workflow begins with block (500) measuring a measured fluid flow rate at a plurality of depths in a well penetrating a hydrocarbon reservoir, in accordance with one or more embodiments. The fluid flow rate may be measured with one or more flowmeters (419).

[0074] In Block 510, a normalized cumulative fluid flow rate for the well is determined based on the measured fluid flow rate at the plurality of depths, in accordance with one or more embodiments. Determining the normalized cumulative fluid flow rate may include determining a first combination of the measured fluid flow rate at the plurality of depths, as shown in Block 512. Determining the first combination may include performing integration with respect to depth, as shown in Block 514. Furthermore, a quotient of the first combination and the measured fluid flow rate at the surface of the well may be determined, as shown in Block 516. The normalized cumulative fluid flow rate may be determined with the left-hand side of Eq. (12).

[0075] In Block 520 a reservoir simulation model for the hydrocarbon reservoir is received, in accordance with one or more embodiments. The reservoir simulation model may include a spatial distribution of porosity and permeability. The reservoir simulation model may include a permeability matrix for a single-porosity model. The reservoir simulation model may also include a fracture permeability. Each of these distributions may be discretized on an array of grid points or an array of grid blocks. The grid blocks may be organized in layers. The reservoir simulation model may include a global fluid flow capacity of the well. The reservoir

simulation model may also include a plurality of layers, and a corresponding plurality of layer fluid flow capacities.

[0076] In Block 530, a normalized fluid flow capacity for the well is determined based on the reservoir simulation model, in accordance with one or more embodiments. Using the distribution of permeability obtained in the previous step, a quotient of a second combination of the plurality of layer fluid flow capacities and the and the global fluid flow capacity of the well may be determined, as shown in Block 532. The normalized fluid flow capacity for the well may be determined using the right-hand side of Eq. (12).

[0077] In Block 540, the reservoir simulation model is updated based, at least in part, on the normalized cumulative fluid flow rate and the normalized fluid flow capacity, in accordance with one or more embodiments. In some embodiments, the reservoir simulation model may include a permeability model. The permeability model may include a matrix permeability. In some embodiments the permeability model includes a matrix permeability and a fracture permeability. The permeability model of the reservoir simulation model may be updated based on a difference measure between the normalized cumulative fluid flow rate and the normalized fluid flow capacity. The permeability model may be updated iteratively, or recursively, until the difference measure is less than a predetermined value.

[0078] In Block 550, a fluid pressure distribution at a first time may be received, where the fluid pressure distribution represents fluid pressure throughout at least a first portion of the hydrocarbon reservoir, in accordance with one or more embodiments. The pore fluid distribution may be discretized on an array of grid points or an array of grid blocks. The pore fluid may be discretized using the same array of grid points as the porosity and permeability or using a distinct but overlapping array of grid points.

[0079] In Block 552, the fluid pressure distribution at a second time may be simulated, where the second time is later than the first time, in accordance with one or more embodiments. Using the input data obtained in previous steps, simulation data may be obtained from a reservoir simulation at a second time. The second time may be, for example, a time during simulated fluid flow. Production data may be simulated as a well flow rate at the second time and pressure may be simulated as the well flowing bottom-hole pressure at the second time. In some embodiments the output pressure data obtained from the reservoir simulation at the second time is the well block pressure. A weighted average pressure over several blocks surrounding the well may be also obtained.

[0080] In Block 554, a location for drilling an infill well may be identified based, at least in part, on the fluid pressure distribution at the second time, in accordance with one or more embodiments. The updated reservoir simulation model may be used in a predictive manner to simulate the fluid flow

dicted at the second time may be used to identify the locations for new infill wells and forecast the potential production of the new wells.

[0081] In Block 556, a wellbore trajectory is planned to intersect the location, in accordance with one or more embodiments. Knowledge of the location of the hydrocarbon reservoir (405) and other formations (424) may then be transferred to the wellbore planning system (110). Instructions associated with the wellbore planning system (110) may be stored, for example, in the memory (1009) within the computer system (1000) described in FIG. 10 below. The wellbore planning system (110) uses the knowledge of the manifestation of the hydrocarbon reservoir (405) and other formations (424) to update a wellbore trajectory (404) within the subterranean region of interest (100). A wellbore trajectory (404) may then be planned to intersect the identified location of new infill wells. The planned wellbore trajectory (404) may be influenced by shallow drilling hazards, such as gas pockets, subterranean water flows, and/or unstable/metastable fault zones.

[0082] In Block 558, the infill well is drilled guided by the planned wellbore trajectory, in accordance with one or more embodiments. The wellbore planning system (110) may transfer the planned wellbore trajectory (404) to the drilling system (400) described in FIG. 4. The drilling system (400) may drill the infill well along the planned wellbore trajectory (404) to access and produce the hydrocarbon reservoir (405) to the surface (124).

[0083] The proposed method is illustrated using numerical examples as is common in the art and is advantageous at least because the solutions may be known a priori. The example starts with a numerical representation of a simplified portion of the earth (the “model”). Then simulated data is generated from the numerical solution of the differential equation describing the physics of fluid flow in porous media. Embodiments of the disclosed invention are applied to the simulated data and the results are compared with the model to assess the performance of the disclosed embodiments.

[0084] In one example, using a reservoir simulator, a constant rate single-flow is simulated on a vertical well traversing the medium of homogeneous permeability (610) shown in FIG. 6. The vertical well fully penetrates the reservoir simulation model composed of 10 layers (612), with layer properties indicated in FIG. 6. Other properties of the reservoir simulation model are detailed in Table 1 below. The simulated fluid flow at the top surface of the well may be represented by a flowmeter response, and is used to determine the normalized cumulative fluid flow rate using the left-hand side of Equation (12). On the other hand, the normalized fluid flow capacity is determined using the values of homogeneous permeability (610) and the right-hand side of Equation (13).

TABLE 1

Qo (stb/d)	Bo (bbl/stb)	μ (cp)	H (ft)	ϕ	Cf (1/psi)	Pi (psi)	Δx and Δy (ft)	Grid size
1500	1.7	0.5	100	0.25	3E(-6)	4800	50	10 × 10 × 10

and pressure within the hydrocarbon reservoir and production from one or more for future reservoir management scenarios. For example, the fluid pressure distribution pre-

[0085] Panel (620) illustrates the comparison of the normalized cumulative fluid flow rate (622) along the depth of the well and the normalized fluid flow capacity at different

layer interfaces (624). The vertical axis (626) indicates the layer indices and the horizontal axis (628) indicates the normalized cumulative fluid flow rate or the normalized fluid flow capacity in percentage. For the constant single-flow and homogeneous permeability model, an excellent agreement can be observed between the normalized cumulative fluid flow rate (622) and the normalized fluid flow capacity at all selected layer interfaces (624).

[0086] In a second example, a constant rate single-flow is simulated on a vertical well traversing the medium of heterogeneous permeability (710) shown in FIG. 7. The reservoir simulation model is also composed of 10 layers (712), with layer properties indicated in FIG. 7. The other properties of the reservoir simulation model are also given in Table 1. Panel (720) illustrates the comparison of the normalized cumulative fluid flow rate (722) along the depth of the well and the normalized fluid flow capacity at different layer interfaces (724). The vertical axis (726) indicates the layer indices and the horizontal axis (728) indicates the normalized cumulative fluid flow rate or the normalized fluid flow capacity in percentage. As seen in Panel (720), in the case of constant single-flow and heterogeneous permeability model, there is also an excellent agreement between the normalized cumulative fluid flow rate (722) and the normalized fluid flow capacity at all selected layer interfaces (724), both obtained with Equation (12).

[0087] Other examples are illustrated in FIGS. 8 and 9 for reservoir simulation models (116) with dual-porosity dual-permeability. The properties of the reservoir simulation model (116) are also given in Table 1. In Panel (810) of FIG. 8 a homogeneous dual-porosity dual-permeability reservoir simulation model is discretized in 10 layers (812), and in panel (910) of FIG. 9 a heterogeneity dual-porosity dual-permeability reservoir simulation model is discretized in 10 layers (912). The vertical axes (826, 926) in Panels (810, 910) indicate the layer indices and the horizontal axes (828, 928) indicate the normalized cumulative fluid flow rate or the normalized fluid flow capacity in percentage. Panels (820, 920) clearly illustrate the very good agreement between the respective normalized cumulative fluid flow rate (822, 922) and the normalized fluid flow capacity at all selected layer interfaces (824, 924), all obtained with Equation (12).

[0088] In some embodiments the wellbore planning system (110), the reservoir simulator (120), and the reservoir interpretation system (480) may each be implemented within the context of a computer system. FIG. 10 is a block diagram of a computer system (1000) used to provide computational functionalities associated with described algorithms, methods, functions, processes, flows, and procedures as described in the instant disclosure, according to an implementation. The illustrated computer (1000) is intended to encompass any computing device such as a high performance computing (HPC) device, a server, desktop computer, laptop/notebook computer, wireless data port, smart phone, personal data assistant (PDA), tablet computing device, one or more processors within these devices, or any other suitable processing device, including both physical or virtual instances (or both) of the computing device. Additionally, the computer (1000) may include a computer that includes an input device, such as a keypad, keyboard, touch screen, or other device that can accept user information, and an output device that conveys information associated with the opera-

tion of the computer (1000), including digital data, visual, or audio information (or a combination of information), or a GUI.

[0089] The computer (1000) can serve in a role as a client, network component, a server, a database or other persistency, or any other component (or a combination of roles) of a computer system for performing the subject matter described in the instant disclosure. The illustrated computer (1000) is communicably coupled with a network (1002). In some implementations, one or more components of the computer (1000) may be configured to operate within environments, including cloud-computing-based, local, global, or other environment (or a combination of environments).

[0090] At a high level, the computer (1000) is an electronic computing device operable to receive, transmit, process, store, or manage data and information associated with the described subject matter. According to some implementations, the computer (1000) may also include or be communicably coupled with an application server, e-mail server, web server, caching server, streaming data server, business intelligence (BI) server, or other server (or a combination of servers).

[0091] The computer (1000) can receive requests over network (1002) from a client application (for example, executing on another computer (1000)) and responding to the received requests by processing the said requests in an appropriate software application. In addition, requests may also be sent to the computer (1000) from internal users (for example, from a command console or by other appropriate access method), external or third-parties, other automated applications, as well as any other appropriate entities, individuals, systems, or computers.

[0092] Each of the components of the computer (1000) can communicate using a system bus (1003). In some implementations, any or all of the components of the computer (1000), both hardware or software (or a combination of hardware and software), may interface with each other or the interface (1004) (or a combination of both) over the system bus (1003) using an application programming interface (API) (1007) or a service layer (1008) (or a combination of the API (1007) and service layer (1008)). The API (1007) may include specifications for routines, data structures, and object classes. The API (1007) may be either computer-language independent or dependent and refer to a complete interface, a single function, or even a set of APIs. The service layer (1008) provides software services to the computer (1000) or other components (whether or not illustrated) that are communicably coupled to the computer (1000). The functionality of the computer (1000) may be accessible for all service consumers using this service layer (1008). Software services, such as those provided by the service layer (1008), provide reusable, defined business functionalities through a defined interface. For example, the interface may be software written in JAVA, C++, or other suitable language providing data in extensible markup language (XML) format or other suitable format. While illustrated as an integrated component of the computer (1000), alternative implementations may illustrate the API (1007) or the service layer (1008) as stand-alone components in relation to other components of the computer (1000) or other components (whether or not illustrated) that are communicably coupled to the computer (1000). Moreover, any or all parts of the API (1007) or the service layer (1008) may be implemented as child or sub-modules of another software

module, enterprise application, or hardware module without departing from the scope of this disclosure.

[0093] The computer (1000) includes an interface (1004). Although illustrated as a single interface (1004) in FIG. 10, two or more interfaces (1004) may be used according to particular needs, desires, or particular implementations of the computer (1000). The interface (1004) is used by the computer (1000) for communicating with other systems in a distributed environment that are connected to the network (1002). Generally, the interface (1004) includes logic encoded in software or hardware (or a combination of software and hardware) and operable to communicate with the network (1002). More specifically, the interface (1004) may include software supporting one or more communication protocols associated with communications such that the network (1002) or interface's hardware is operable to communicate physical signals within and outside of the illustrated computer (1000).

[0094] The computer (1000) includes at least one computer processor (1005). Although illustrated as a single computer processor (1005) in FIG. 10, two or more processors may be used according to particular needs, desires, or particular implementations of the computer (1000). Generally, the computer processor (1005) executes instructions and manipulates data to perform the operations of the computer (1000) and any algorithms, methods, functions, processes, flows, and procedures as described in the instant disclosure.

[0095] The computer (1000) also includes a memory (1009) that holds data for the computer (1000) or other components (or a combination of both) that may be connected to the network (1002). For example, memory (1009) may be a database storing data consistent with this disclosure. Although illustrated as a single memory (1009) in FIG. 10, two or more memories may be used according to particular needs, desires, or particular implementations of the computer (1000) and the described functionality. While memory (1009) is illustrated as an integral component of the computer (1000), in alternative implementations, memory (1009) may be external to the computer (1000).

[0096] The application (1006) is an algorithmic software engine providing functionality according to particular needs, desires, or particular implementations of the computer (1000), particularly with respect to functionality described in this disclosure. For example, application (1006) can serve as one or more components, modules, applications, etc. Further, although illustrated as a single application (1006), the application (1006) may be implemented as multiple applications (1006) on the computer (1000). In addition, although illustrated as integral to the computer (1000), in alternative implementations, the application (1006) may be external to the computer (1000).

[0097] There may be any number of computers (1000) associated with, or external to, a computer system containing computer (1000), each computer (1000) communicating over network (1002). Further, the term "client," "user," and other appropriate terminology may be used interchangeably as appropriate without departing from the scope of this disclosure. Moreover, this disclosure contemplates that many users may use one computer (1000), or that one user may use multiple computers (1000).

[0098] In some embodiments, the computer (1000) is implemented as part of a cloud computing system. For example, a cloud computing system may include one or

more remote servers along with various other cloud components, such as cloud storage units and edge servers. In particular, a cloud computing system may perform one or more computing operations without direct active management by a user device or local computer system. As such, a cloud computing system may have different functions distributed over multiple locations from a central server, which may be performed using one or more Internet connections. More specifically, cloud computing system may operate according to one or more service models, such as infrastructure as a service (IaaS), platform as a service (PaaS), software as a service (SaaS), mobile "backend" as a service (MBaaS), serverless computing, artificial intelligence (AI) as a service (AIaaS), and/or function as a service (FaaS).

[0099] Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims.

1. A method comprising:

measuring, using a flowmeter, a measured fluid flow rate at a plurality of depths in a well penetrating a hydrocarbon reservoir;

using a reservoir simulator:

determining a normalized cumulative fluid flow rate for the well based on the measured fluid flow rate at the plurality of depths;

receiving a reservoir simulation model for the hydrocarbon reservoir;

determining a normalized fluid flow capacity for the well based on the reservoir simulation model; and

updating the reservoir simulation model based, at least in part, on the normalized cumulative fluid flow rate and the normalized fluid flow capacity.

2. The method of claim 1, wherein the reservoir simulation model comprises a permeability matrix.

3. The method of claim 1, wherein the reservoir simulation model comprises a fracture permeability.

4. The method of claim 1, wherein determining the normalized cumulative fluid flow rate comprises determining a first combination of the measured fluid flow rate at the plurality of depths.

5. The method of claim 4, wherein

the measured fluid flow rate at the plurality of depths comprises a measured fluid flow rate at a surface of the well; and

determining the normalized cumulative fluid flow rate further comprises determining a quotient of the first combination and the measured fluid flow rate at the surface of the well.

6. The method of claim 1, wherein:

the reservoir simulation model comprises a fluid flow capacity of the well, a plurality of layers, and a corresponding plurality of layer fluid flow capacities; and

determining the normalized fluid flow capacity comprises determining a quotient of a second combination of the plurality of layer fluid flow capacities and the fluid flow capacity of the well.

7. The method of claim 5, wherein determining the first combination comprises performing integration with respect to depth.

8. The method of claim 1, wherein updating the reservoir simulation model is based, at least in part, on a difference between the normalized cumulative fluid flow rate and the normalized fluid flow capacity.

9. The method of claim 1, further comprising, receiving, using the reservoir simulator, a fluid pressure distribution at a first time, wherein the fluid pressure distribution represents fluid pressure throughout at least a first portion of the hydrocarbon reservoir; and simulating, using the reservoir simulator and the updated reservoir simulation model, the fluid pressure distribution at a second time, wherein the second time is later than the first time.

10. The method of claim 9, further comprising:

identifying, using a reservoir interpretation system, a location for drilling an infill well based, at least in part, on the fluid pressure distribution at the second time; planning, using a wellbore planning system, a wellbore trajectory to intersect the location; and drilling, using a drilling system, the infill well guided by the planned wellbore trajectory.

11. A system comprising:

a flowmeter configured to measure a measured fluid flow rate at a plurality of depths in a well penetrating a hydrocarbon reservoir; and

a reservoir simulator configured to receive the measured fluid flow rate at a plurality of depths and to: determine a normalized cumulative fluid flow rate for the well based on the measured fluid flow rate at the plurality of depths,

receive a reservoir simulation model for the hydrocarbon reservoir,

determine a normalized fluid flow capacity for the well based on the reservoir simulation model, and

update the reservoir simulation model based, at least in part, on the normalized cumulative fluid flow rate and the normalized fluid flow capacity.

12. The system of claim 11, wherein the reservoir simulation model comprises a permeability matrix.

13. The system of claim 11, wherein the reservoir simulation model comprises a fracture permeability.

14. The system of claim 11, wherein:

the reservoir simulation model comprises a fluid flow capacity of the well, a plurality of layers, and a corresponding plurality of layer fluid flow capacities, and

the reservoir simulator is further configured to determine the normalized fluid flow capacity based on a quotient of a first combination of the plurality of layer fluid flow capacities and the fluid flow capacity of the well.

15. The system of claim 11, wherein the reservoir simulator is further configured to determine the normalized cumulative fluid flow rate based, at least in part, on a second combination of the measured fluid flow rate at the plurality of depths.

16. The system of claim 15, wherein the reservoir simulator is further configured to determine the second combination by performing integration with respect to depth.

17. The system of claim 15, wherein

the measured fluid flow rate at the plurality of depths comprises a measured fluid flow rate at a surface of the well; and

the reservoir simulator is further configured to determine the normalized cumulative fluid flow rate based on a quotient between the second combination and the measured fluid flow rate at the surface of the well.

18. The system of claim 11, wherein the reservoir simulator is further configured to update the reservoir simulation model based, at least in part, on a difference between the normalized cumulative fluid flow rate and the normalized fluid flow capacity.

19. The system of claim 11, wherein the reservoir simulator is further configured to:

receive a fluid pressure distribution at a first time, wherein the fluid pressure distribution represents fluid pressure throughout at least a first portion of the hydrocarbon reservoir; and

simulate, using the updated reservoir simulation model, the fluid pressure distribution at a second time.

20. The system of claim 19, further comprising:

a reservoir interpretation system configured to identify a location for drilling an infill well based, at least in part, on the fluid pressure distribution at the second time;

a wellbore planning system configured to plan a wellbore trajectory to intersect the location; and

a drilling system configured to drill the infill well guided by the planned wellbore trajectory.

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